

Presynaptic nerve terminals release neurotransmitters via synaptic vesicle exocytosis. This action is mediated by the membrane-bound target-SNARE (t-SNARE) proteins syntaxin and SNAP-25, and the vesicle-associated SNARE (v-SNARE) protein synaptobrevin. Membrane fusion is initiated when the t-SNAREs and v-SNAREs form an α -helix bundle known as the core-*trans*-complex, achieved by zipping SNARE proteins together to form a more stable *cis*-complex (Figure 1).

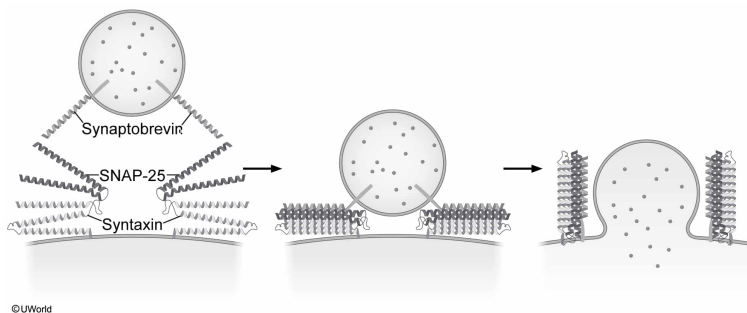


Figure 1 Formation of the core-*trans*-complex and subsequent membrane fusion

The *cis*-complex contains a region called the zero ionic layer, which is the main site of interaction between complexed proteins. Within the zero ionic layer, arginine on synaptobrevin coordinates with carbonyl groups present on the other residues shown in Table 1. The zero ionic layer is buried within amphipathic domains called leucine zippers, which shield the zero ionic layer from solvent molecules.

Table 1 Key Amino Acids of the *cis*-Complex Zero Ionic Layer

SNARE protein	Zero ionic layer residue
Synaptobrevin	R56
Syntaxin	Q226
SNAP-25 (Sn1)	Q53

Once membrane fusion occurs, the *cis*-complex must disassemble. An early step in *cis*-complex disassembly is destabilization by the ATPase *N*-ethylmaleimide-sensitive factor (NSF), which disrupts leucine zipper regions, leading to vesicle reuptake. Movement of NSF to the cell membrane is mediated by the cytoplasmic protein α -SNAP, which recruits NSF after binding the N-terminal domain of syntaxin.

To determine the effect of synaptobrevin on the binding of α -SNAP to syntaxin, increasing amounts of recombinant His-tagged α -SNAP were added to a constant amount of syntaxin affixed to glutathione-agarose beads in the presence or absence of synaptobrevin. After incubation, bound α -SNAP was recovered and quantitatively analyzed by immunoblotting. Results are shown in Figure 2.

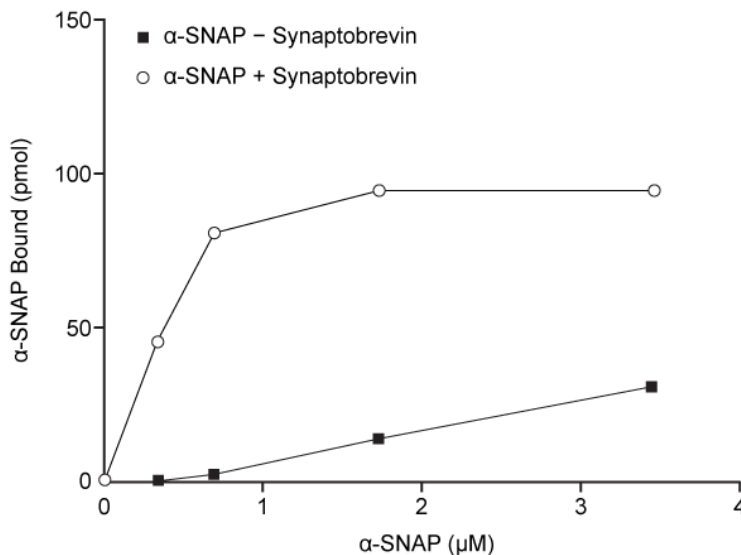


Figure 2 α -SNAP binding to syntaxin in the absence (square) or presence (circle) of synaptobrevin

During *cis*-complex disassembly, what consequence of NSF function is most likely responsible for destabilization of the *cis*-complex?

- ✔ ☒ A. Exposure of the zero ionic layer's charged amino acids to

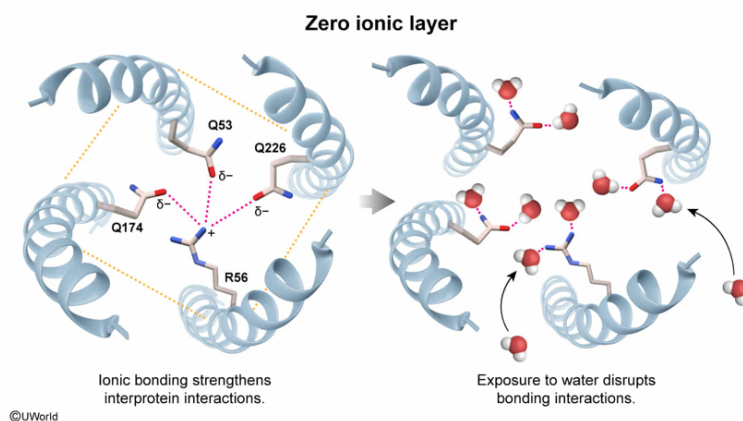
solvent molecules [66%]

- ☐ B. Increased repulsive interactions among charged residues in the zero ionic layer [19%]
- ☐ C. Increased amphipathic qualities of the leucine zipper dimer [10%]
- ☐ D. Release of v-SNARE proteins from synaptic vesicles [4%]

Correct

66% answered correctly

Explanation:



Typically, **hydrophilic amino acids** (polar and charged) are found on the **outside of proteins**, where they interact favorably with water molecules. **Hydrophobic amino acids** are usually shielded from the solvation layer in the **protein interior** or found at **protein-protein interfaces**. This arrangement is entropically favorable, as water molecules cannot form hydrogen bonds with hydrophobic residues, and therefore must organize around these residues.

Leucine is hydrophobic and **leucine zipper** layers act as a watertight seal around the zero ionic layer, in which a positively charged, basic **arginine (R) residue** interacts with polar **glutamine (Q) residues** at the interface of the complexed proteins.

NSF destabilizes the *cis*-complex by breaking the leucine zipper seal (ie, unzipping). Without the watertight seal provided by the leucine

zipper, the charged and polar residues of the zero ionic layer are exposed to water. Because water is polar, exposure of the charged and polar residues results in **preferential binding to water** and resultant **destabilization** of the *cis*-complex.

(Choice B) The charges on residues in the zero ionic layer do not change, so no additional repulsive forces are added by breaking the leucine zipper.

(Choice C) A leucine zipper is a common protein motif composed of two amphipathic α -helices, meaning they possess both hydrophilic and hydrophobic residues. Unzipping does not change the residues, and therefore would not change the amphipathic qualities of the zipper.

(Choice D) Release of v-SNARE proteins does not occur until after *cis*-complex disassembly.

Educational objective:

Hydrophilic amino acids are typically found on the outside of proteins whereas hydrophobic amino acids exist in the protein interior and protein-protein interfaces. When polar or charged amino acids are exposed to water, they preferentially form hydrogen bonds with water molecules.

Time spent:0 sec

Subject:
Biochemistry

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Amino Acids and Proteins